SPARQL, a query language for RDF databases

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Computational Management of Data – Part II (A.Y. 2025/2026)
Second Cycle Degree in Digital Humanities and Digital Knowledge
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Summary of the previous lectures (1/4)

A datum is a declarative statement subject-predicate-object that, through the predicate, either **attributes** a **literal** (i.e. a value such as a string, a number, etc.) to a subject **entity** *or* it **relates** such a subject **entity** with another **entity**

Each entity, being used either as subject or object of a statement, is characterised by **a unique identifier**

The **same entity** can be used as subject or object in one or more data, while a literal **cannot be used** as subject in any datum

An attribute is intrinsically **part of** the entity to which it is associated – modifying the value of an attribute affect **only** the entity to which it refers to

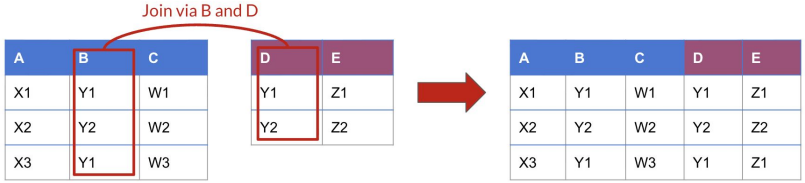
A **data model** is an abstract, simplified and formal representation of some data related to a system or a real domain, and enables us to describe what a data collection is about and to check data correctness

A **data model** permit one to specify <u>classes of entities, their attributes and relations</u>

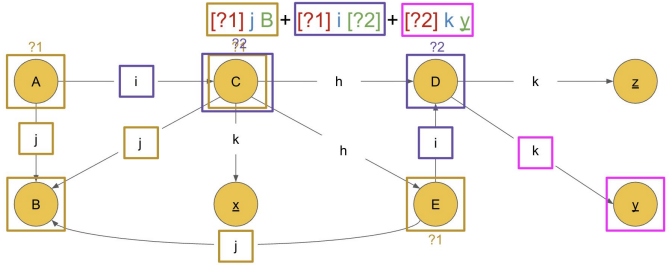
Summary of the previous lectures (2/4)

Depending on the structure in which data are stored , queries to datasets are approached differently:

- With **tabular data**, often you have to combine tables between them to obtain bigger tables which contain the query requirements and the related answer



- With **graph data**, you explore the graph starting from fixed points (i.e. known entities, values, predicates) to find a pattern that is compliant with the query



We use **descriptive statistics** as methods to summarize and describe data in a quantitative way – (measures of central tendency) Mean; Median; Mode, (measures of variability) minimum, maximum, and standard deviation

Summary of the previous lectures (3/4)

The creation of the database structures, the mechanism to upload new data into the database, and all the facilities for querying database data, are enabled by the **database management system (DBMS) –** available as **standalone** applications or Embedded DBMS

A **transaction** is a unit of work performed within a DBMS against a database – it represents any change in a database – are one of the main components that enables the integrity of data, being them executed fully or causing a rollback of the database

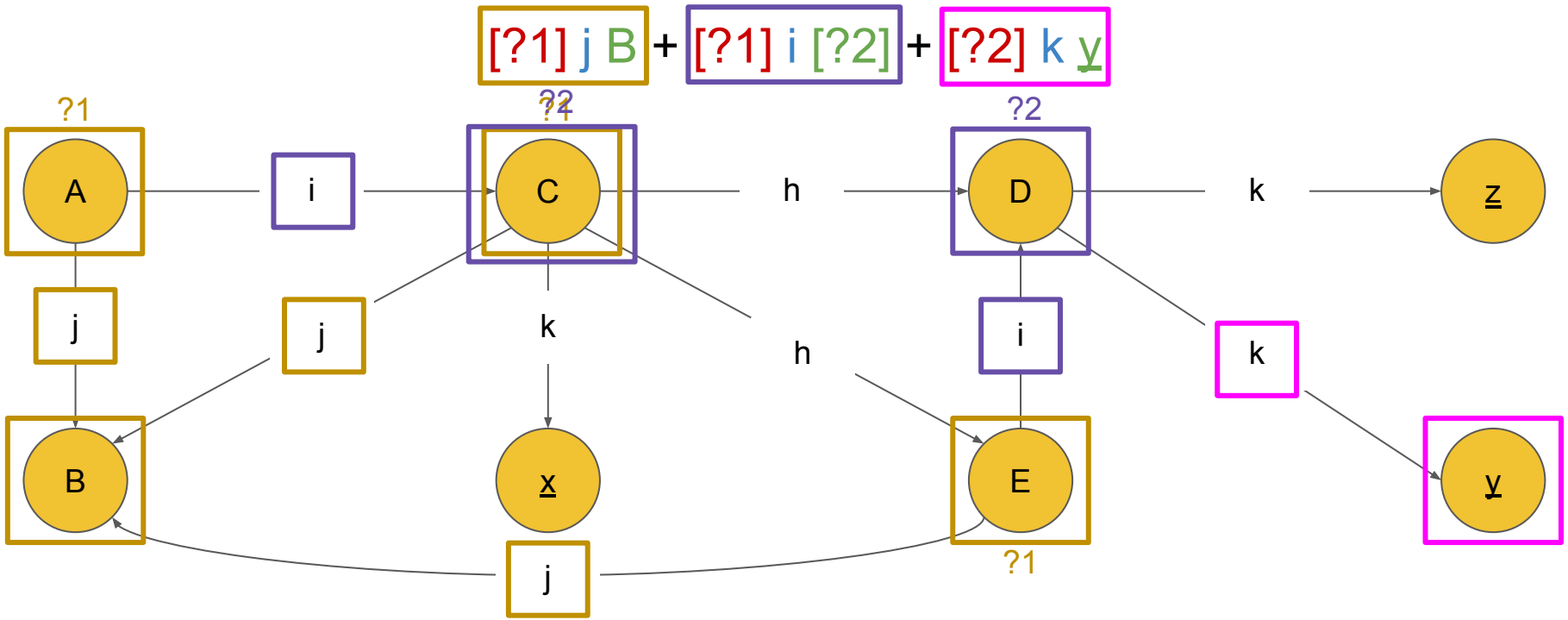
Summary of the previous lectures (4/4)

`SELECT <columns>`	A comma-separated list of columns to have as result of the query ("*" means *all the columns*)
`FROM <tables>`	The table(s) from which to retrieve the data
`WHERE <conditions>`	*[OPTIONAL]* A condition that must hold to include the row of a table in the result of the query
`;`	

Questions to remind for designing a query:

- Which tables do you need?
- Which columns do you select?

A quick recap about querying graphs



SPARQL

The SPARQL Protocol and RDF Query Language (SPARQL, and yes: it is a recursive name) is a query language used and designed for managing RDF data, as those available in an RDF triplestore

It is a standard that is implemented in all RDF-based triplestore, and allows one to populate and query them using a particular syntax

The full specification of the language is a W3C Recommendation and it is entirely available at https://www.w3.org/TR/sparql11-query/

A good tutorial is available at: https://docs.data.world/tutorials/sparql/primer_intro.html

Basic SPARQL syntax for queries

```
SELECT <variables>

WHERE {

  <triple pattern 1> .
  <triple pattern 2> .
  …
}
```

A space-separated list of **`?x`** variables to have as result of the query ("*" means *all the variables*)

A series of conditions involving the **`?x`** variables above used in one or more…

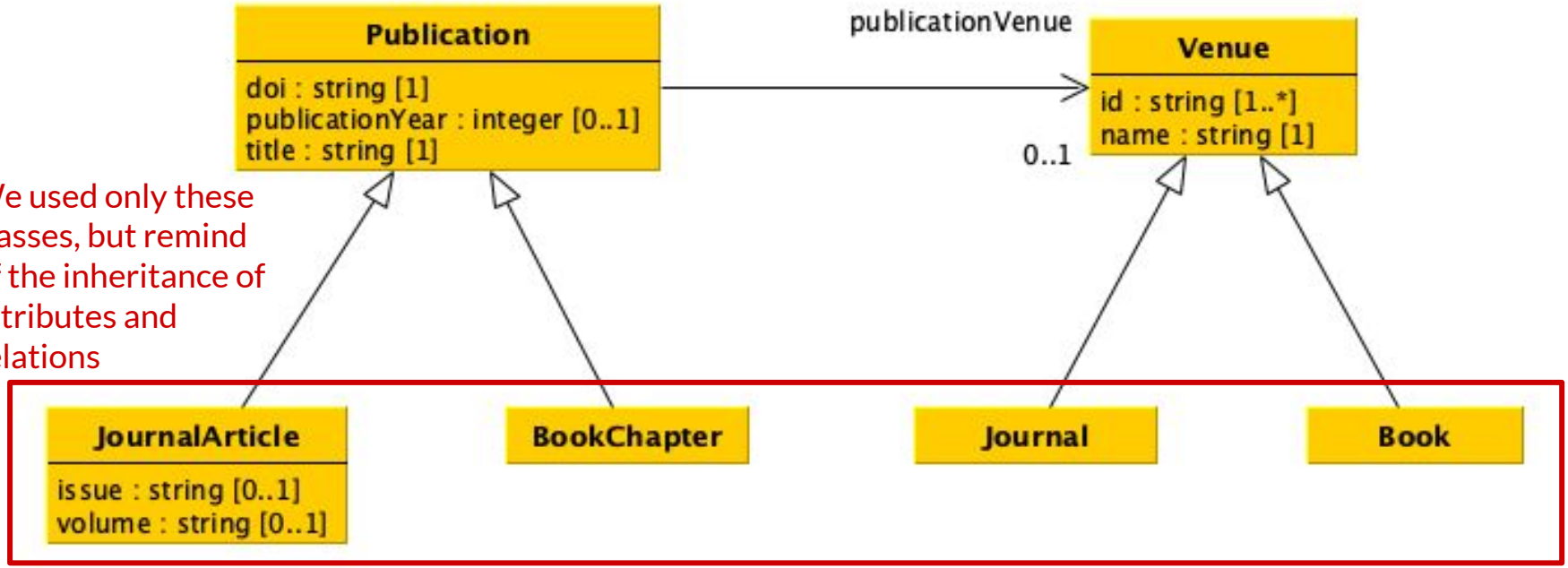
… triple patterns **`S P O`** where:

- **`S`** can be a **`<url>`** resource or a **`?x`** variable
- **`P`** can be a **`<url>`** property or a **`?x`** variable
- **`O`** can be a **`<url>`** resource, a **`"literal"`** or a **`?x`** variable

Questions to remind for designing a query:

- Which resources are known?
- Which literals are known?
- Which properties are known?

Exemplar classes and properties for queries



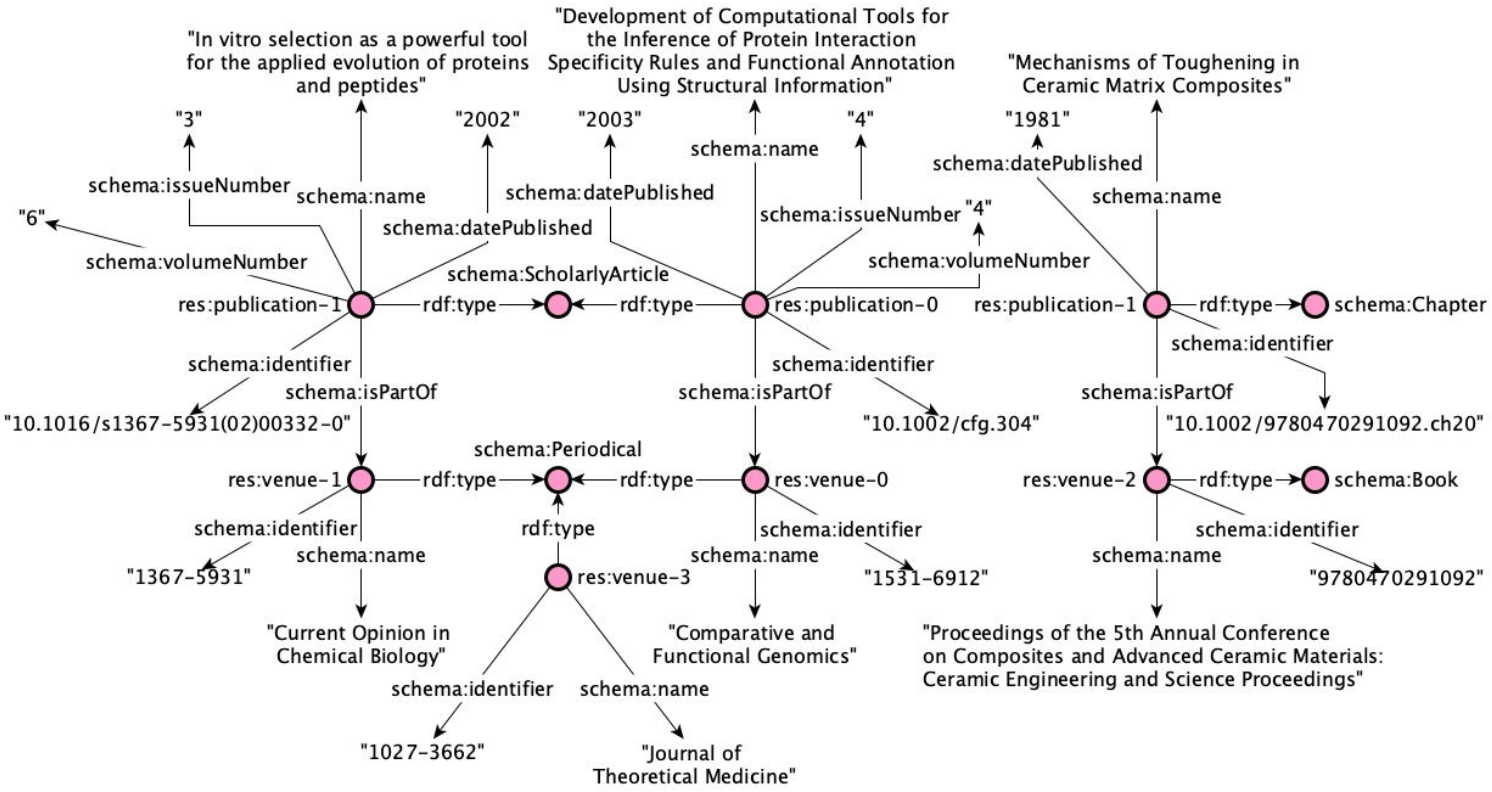
We used only these classes, but remind of the inheritance of attributes and relations

Remind that we used strings to store years of publication instead of integers

Data in the RDF graph

Prefixes

```
PREFIX res:    <https://comp-data.github.io/res/>
PREFIX rdf:    <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX schema: <https://schema.org/>
```



Queries

(Q1) Retrieve complete information about all journal articles

(Q2) Retrieve the titles of all journal articles

(Q3) Retrieve the title of the journal article with DOI “10.1016/s1367-5931(02)00332-0”

(Q4) Retrieve the title of the publication with DOI “10.1016/s1367-5931(02)00332-0”

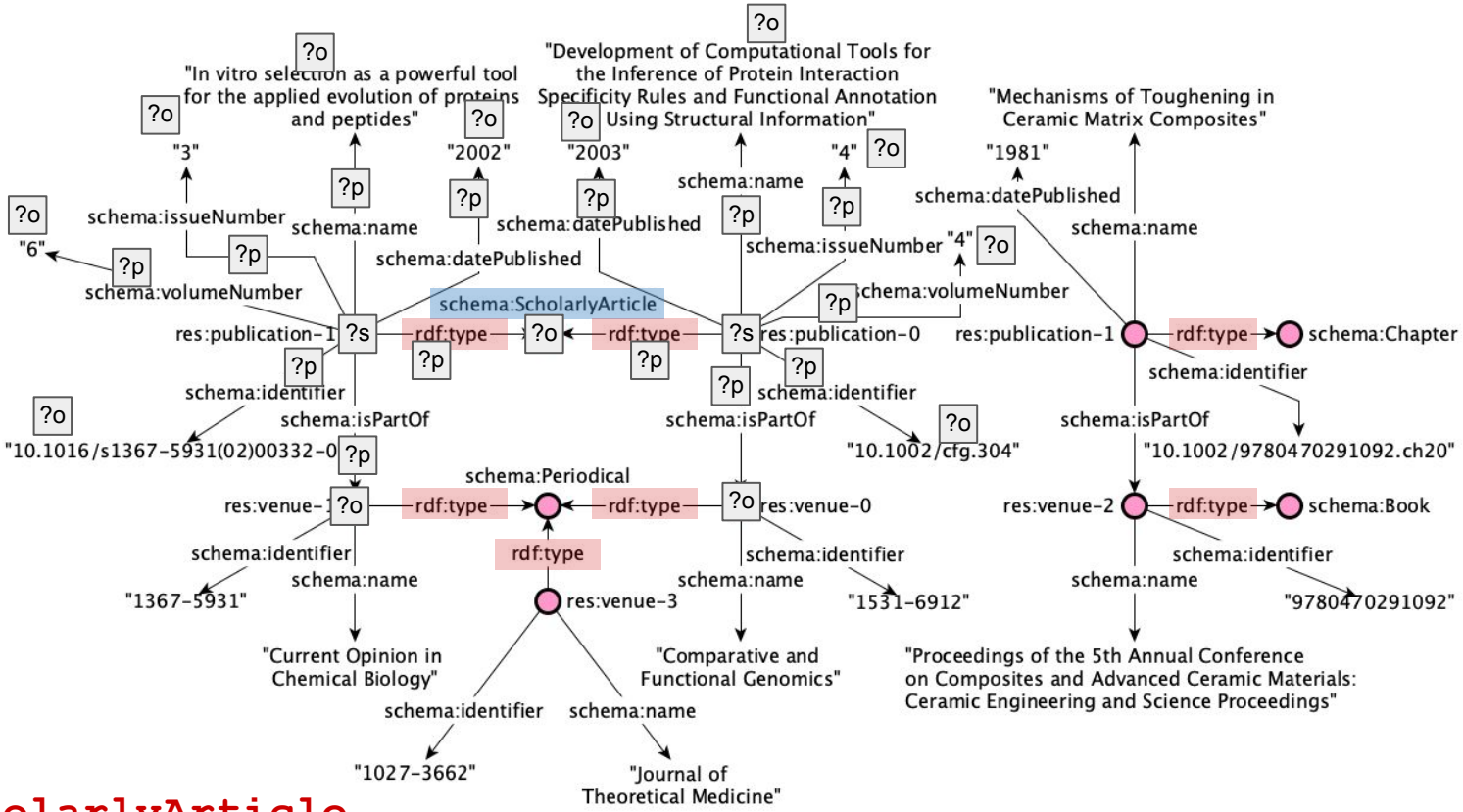
(Q5) Return the name of the journal of the article with DOI “10.1016/s1367-5931(02)00332-0”

(Q6) Return the id and name of the journal of the article with DOI “10.1016/s1367-5931(02)00332-0”

Query 1

Retrieve complete information about all journal articles

- Which resources are known?
- Which literals are known?
- Which properties are known?

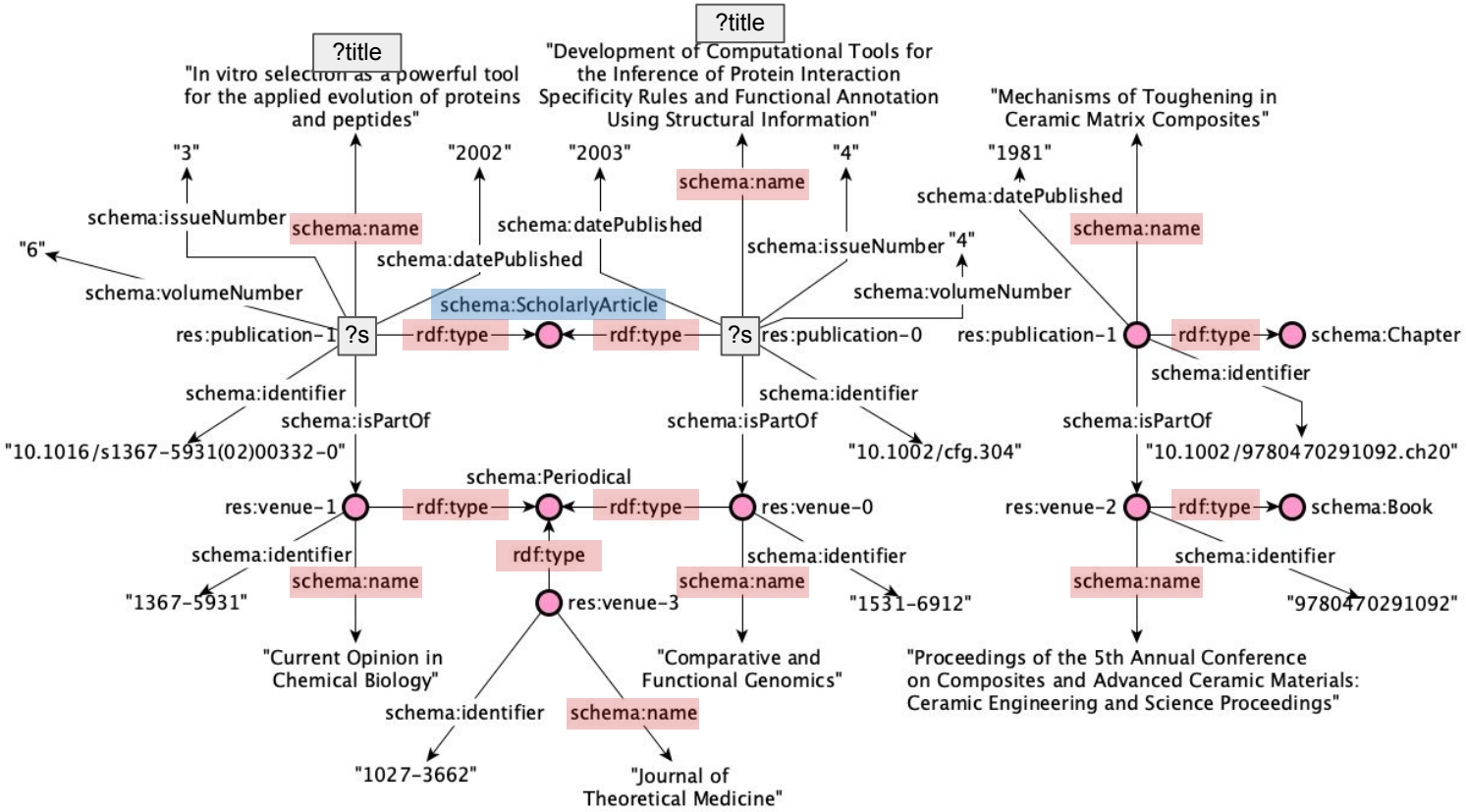



```
SELECT *
WHERE {
  ?s rdf:type
     schema:ScholarlyArticle .
  ?s ?p ?o .
}
```

Query 2

Retrieve the titles of all journal articles

- Which resources are known?
- Which literals are known?
- Which properties are known?

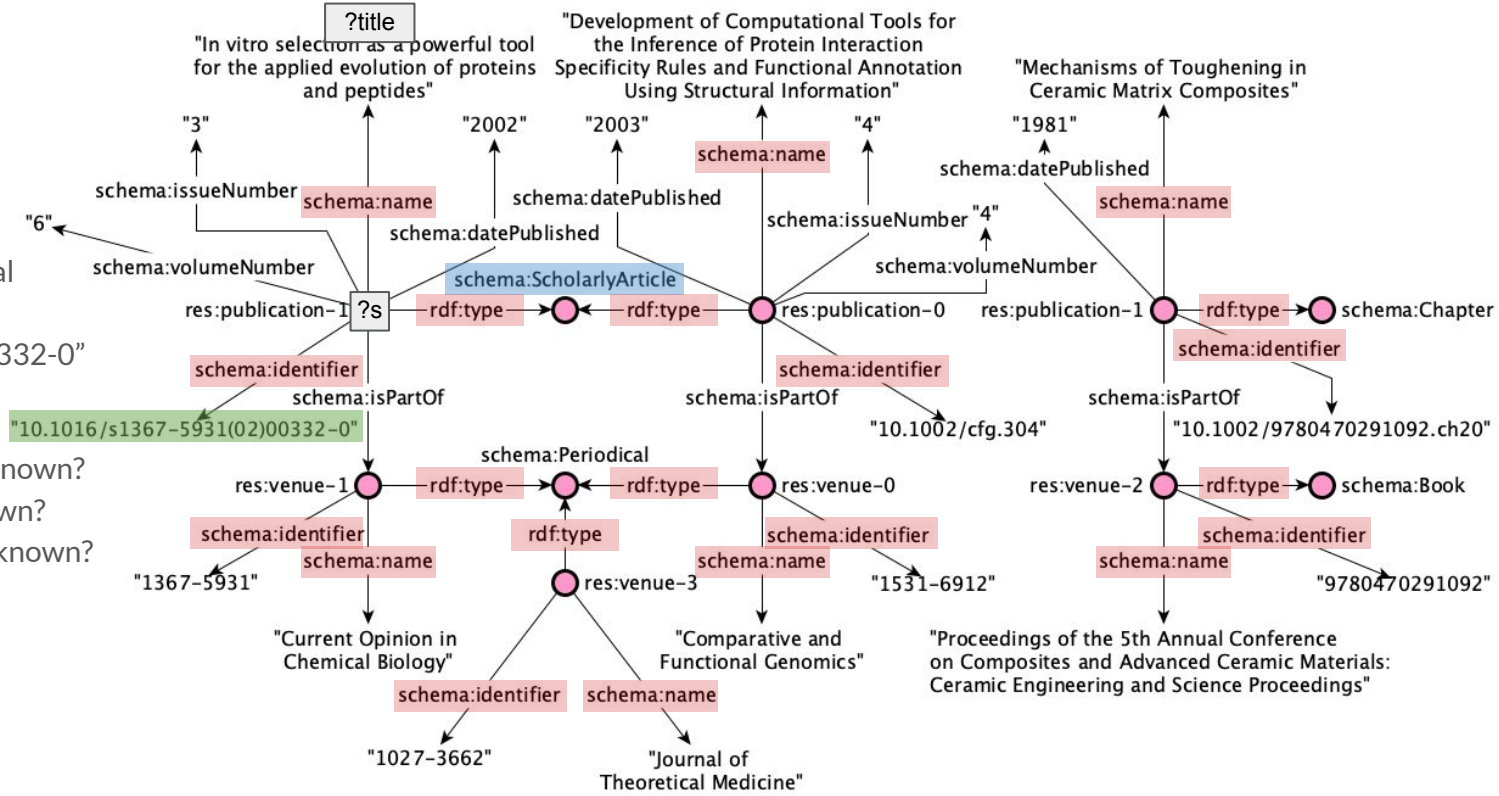



```
SELECT ?title
WHERE {
  ?s rdf:type
     schema:ScholarlyArticle .
  ?s schema:name ?title .
}
```

Query 3

Retrieve the title of the journal article with DOI "10.1016/s1367-5931(02)00332-0"

- Which resources are known?
- Which literals are known?
- Which properties are known?

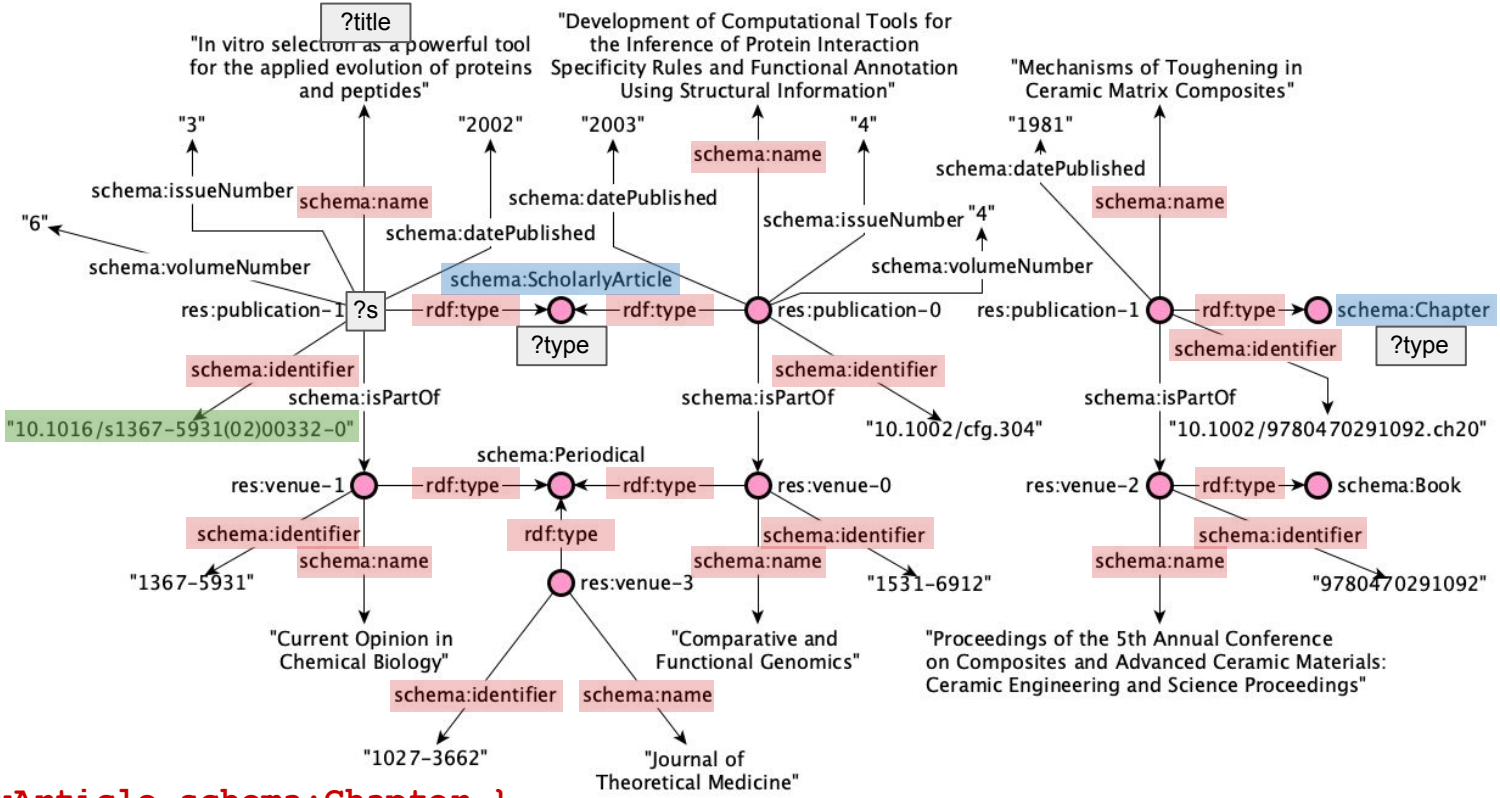



```
SELECT ?title
WHERE {
  ?s rdf:type
     schema:ScholarlyArticle .
  ?s schema:name ?title .
  ?s schema:identifier "10.1016/s1367-5931(02)00332-0" .
}
```

Query 4

Retrieve the title of the publication with DOI "10.1016/s1367-5931(02)00332-0"

- Which resources are known?
- Which literals are known?
- Which properties are known?

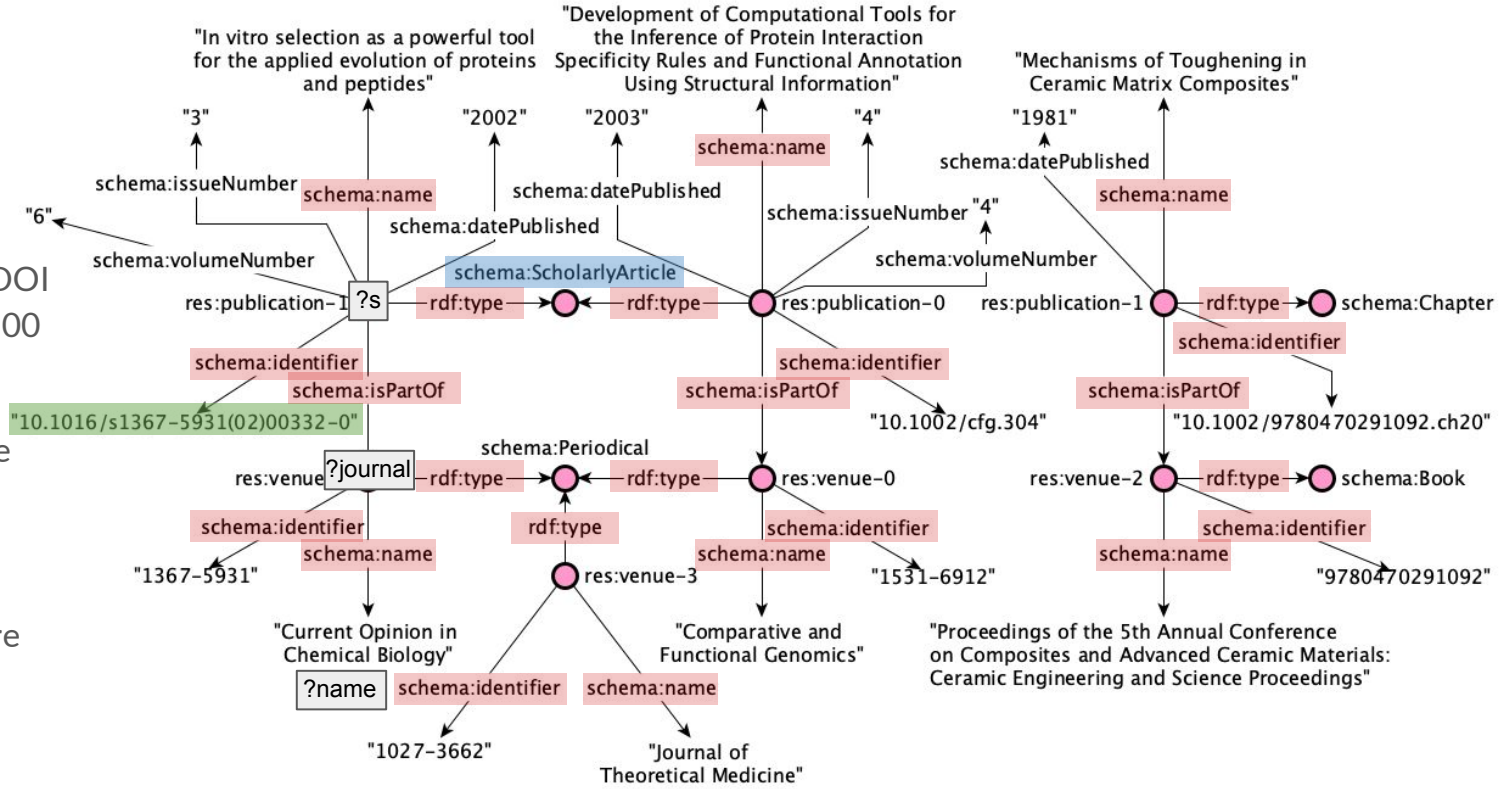



```
SELECT ?title
WHERE {
  VALUES ?type {
    schema:ScholarlyArticle schema:Chapter }
  ?s rdf:type ?type .
  ?s schema:name ?title .
  ?s schema:identifier "10.1016/s1367-5931(02)00332-0" . }
```

Query 5

Return the name of the journal of the article with DOI "10.1016/s1367-5931(02)00332-0"

- Which resources are known?
- Which literals are known?
- Which properties are known?

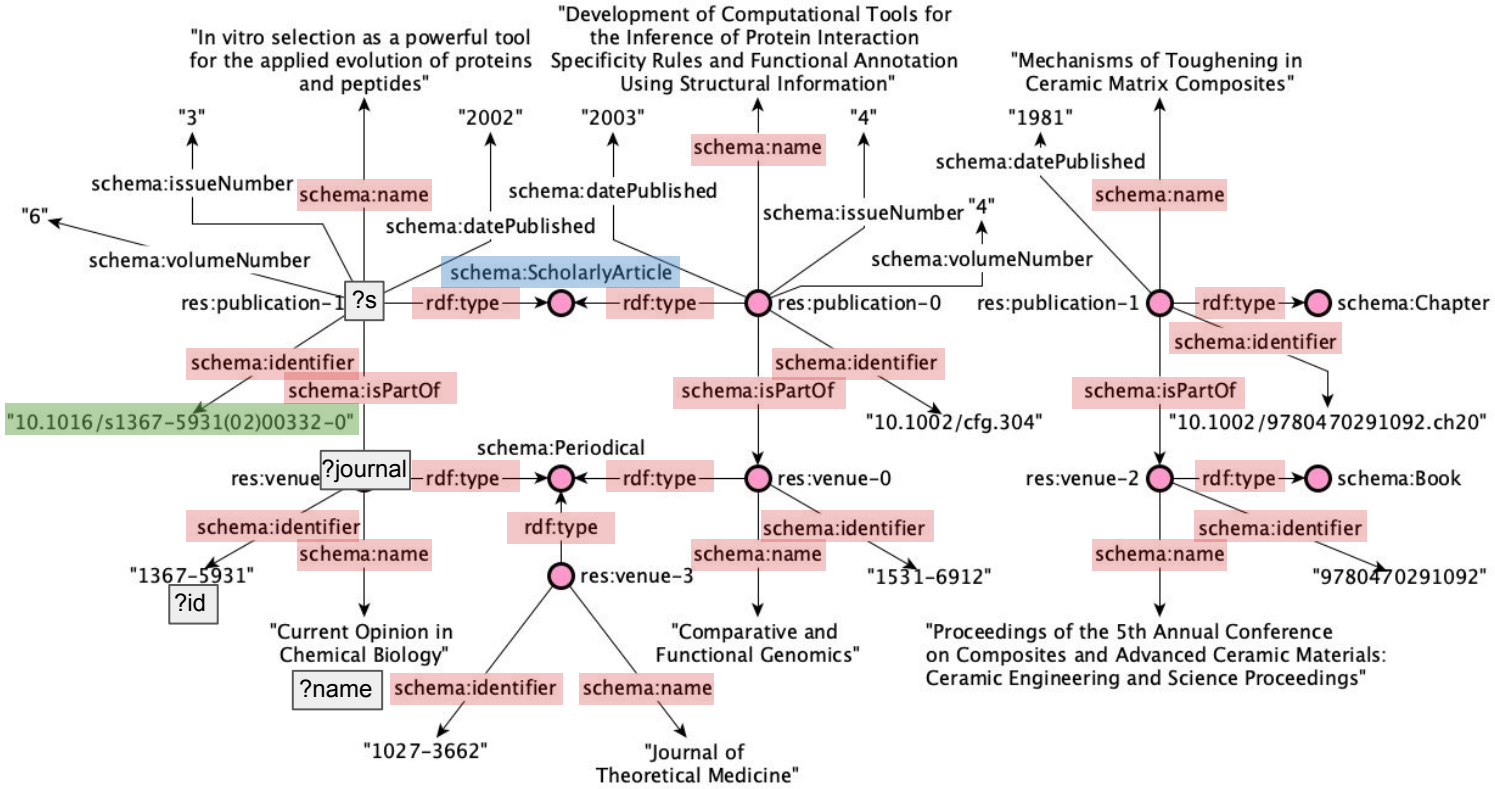



```
SELECT ?name
WHERE {
  ?s rdf:type schema:ScholarlyArticle .
  ?s schema:identifier "10.1016/s1367-5931(02)00332-0" .
  ?s schema:isPartOf ?journal .
  ?journal schema:name ?name . }
```

Query 6

Return the id and name of the journal of the article with DOI “10.1016/s1367-5931(02)00332-0”

- Which resources are known?
- Which literals are known?
- Which properties are known?



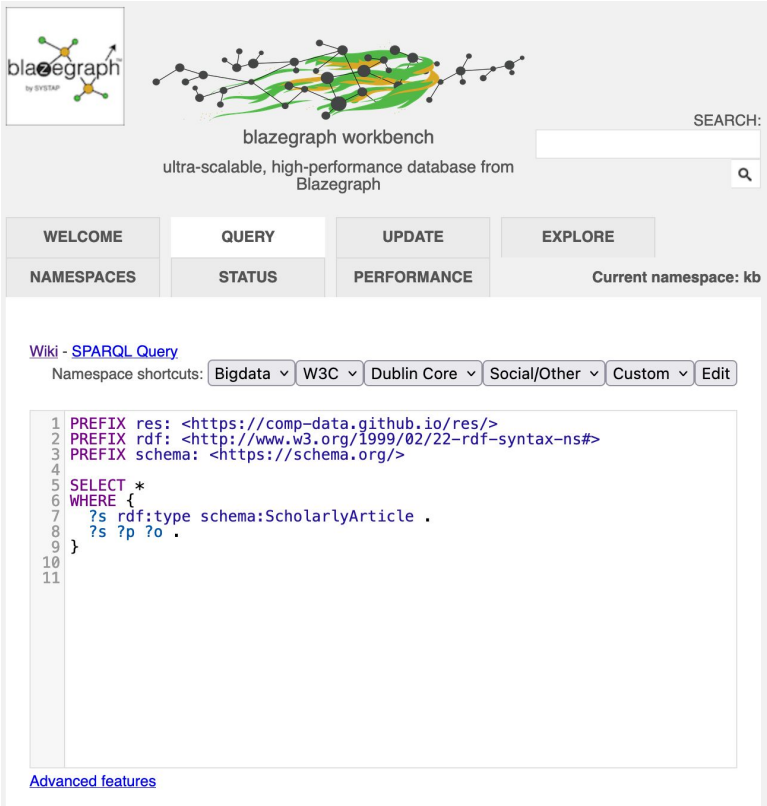

```
SELECT ?name ?id
WHERE {
  ?s rdf:type schema:ScholarlyArticle .
  ?s schema:identifier "10.1016/s1367-5931(02)00332-0" .
  ?s schema:isPartOf ?journal .
  ?journal schema:name ?name .
  ?journal schema:identifier ?id . }
```

Try this on Blazegraph

Download and Run Blazegraph following: https://thinkcompute.github.io/20-graph-database.html#blazegraph-a-database-for-rdf-data

Open your browser and go to **http://127.0.0.1:9999/blazegraph/**

Select the tab “Query” and write and execute SPARQL queries (remember the prefixes if you want to use them in the query)



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